אוניברסיטת
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בשומרון

Advanced Quantization Techniques for Accelerating Machine Learning

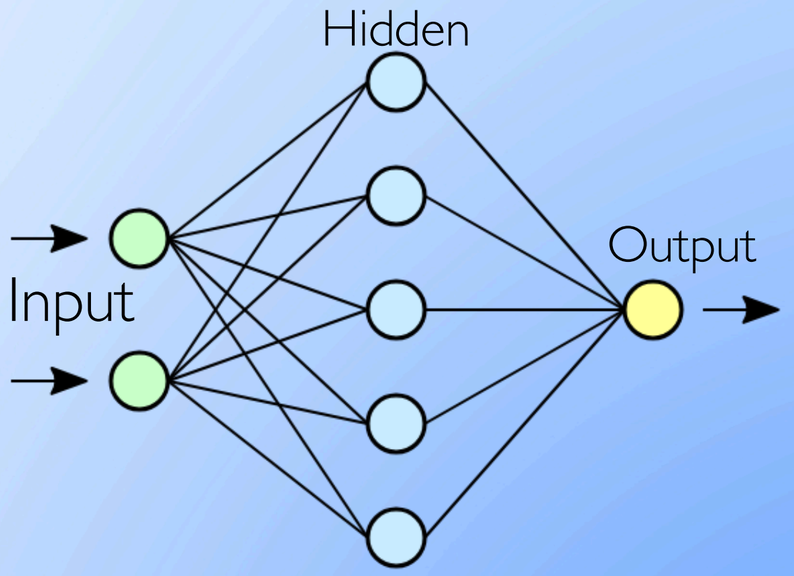
Mentor: Itamar Cohen
Presenters: Noam Cohen, Yair Horvits, Bar Yechezkel | 2025

Ariel University | Computer Science Department

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Motivation – Accelerating NN
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What is Dequantization?
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int8 or int16 Precision
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Quantized Equation
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Matrix Multiplication in Neural Networks

Neural Networks- Background

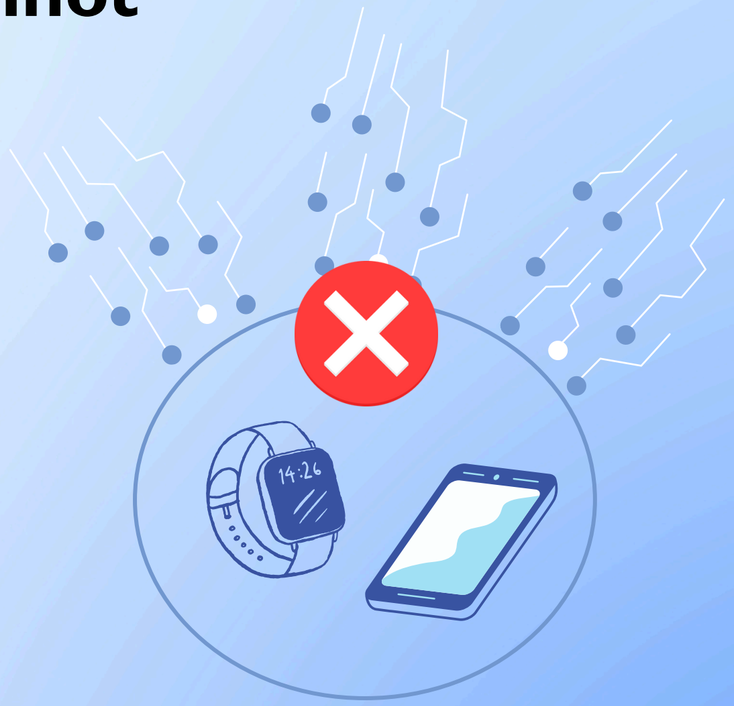
- Neural networks are made of layers of neurons, and each connection between neurons has a weight.
- More layers/neuron → more **parameters** → more **computations** → more **memory & energy usage**.
- Large networks achieve higher **accuracy**, but at the cost of **heavy resource requirements**.



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Motivation – Accelerating NN

- Modern neural networks can have billions of parameters.
- Running them in full precision is **slow, memory-intensive, and power-hungry.**
- Edge devices like phones and IoT devices **cannot** efficiently run large models.

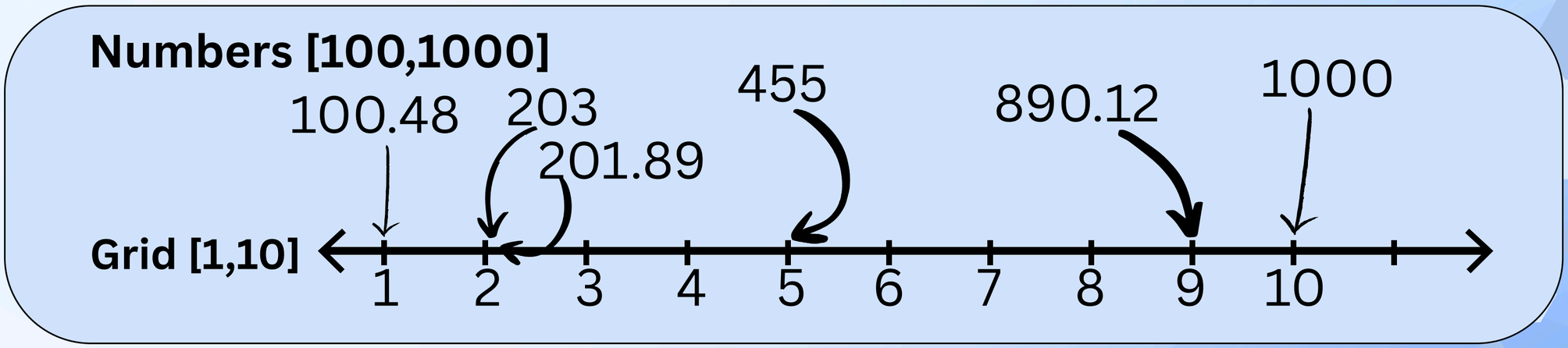


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What is Quantization?

Definition: Quantization is the process of constraining an input from a continuous or otherwise large set of values (such as the real numbers) to a discrete set (such as the integers).

Example:

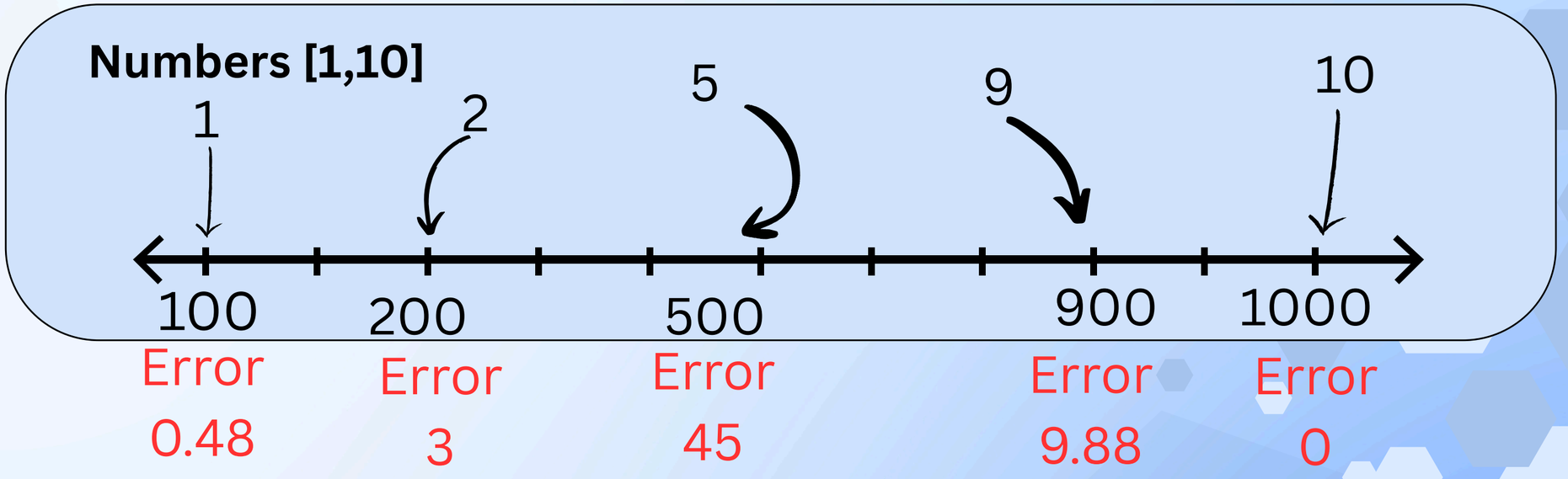


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Results -Accuarcy

What is Dequantization?

Dequantization is the process of mapping quantized values back to approximate continuous values.

The reconstructed values are close to the original, but not exact, so small errors may occur



Sum of Errors
58.36!

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Quantization: Benefits and Trade-offs

Benefits

- Reduces model size (smaller memory footprint)
- Speeds up inference (faster computation, lower latency)
- Enables deployment on edge and mobile devices

Trade-offs

- Reconstruction error during dequantization
- Quantization and dequantization must be applied at every layer , which adds latency and power consumption.

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Quantization Methods in Literature

1

Post-Training Quantization (PTQ)

- Apply quantization after training a full-precision model
- Very quick and simple, no retraining needed
- Slight accuracy drop (model not optimized for lower precision)

2

Quantization-Aware Training (QAT)

- Integrates quantization during training
- Helps the model adapt and keep accuracy high
- Best for complex tasks
- Drawback: longer training time, more resources

In this project, we focus on **PTQ**, since it's simple, fast, and practical for real deployment

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Goal of Our Work

Main Goal: Find the quantization method that maximizes accuracy while minimizing dimension reduction impact.

We explore different strategies:

1. **Symmetric** vs. **Asymmetric** quantization
2. **INT8** vs. **INT16** precision
3. **Clamping** strategies for outliers

(values that deviate significantly from the rest)

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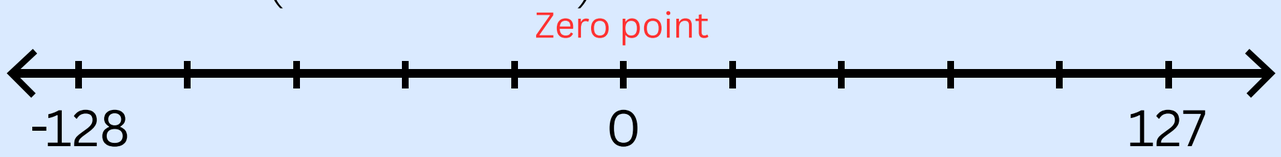
Symmetric VS Asymmetric

Symmetric:

Uses a balanced range centered at zero for positive and negative values.

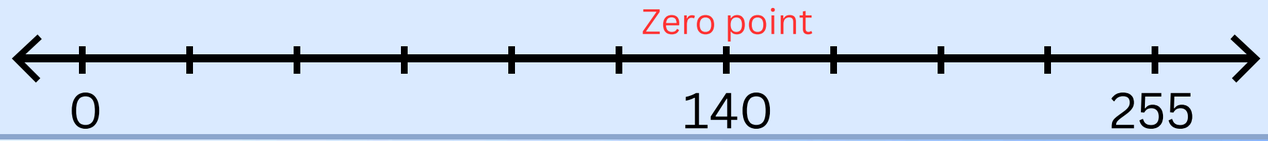
Ideal for data evenly distributed around zero, like many neural network weights.

Example: For 8-bit integers, range is -128 to 127 (zero-centered).



Asymmetric:

- Moves the zero point to match the data's pattern, not fixed at zero.
- Best for data that isn't balanced, like activations with mostly negative values
- Example: For 8-bit integers, range is 0 to 255, with zero point at 140 for negative-heavy data.

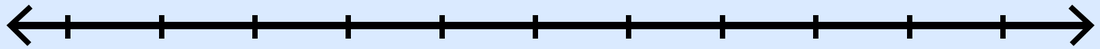


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int8 or int16 Precision

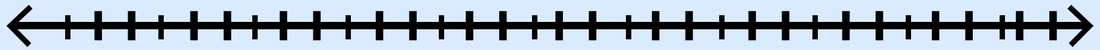
int8

- Uses 8 bits, storing 256 possible values.
- Example: Range can be -128 to 127 (signed) or 0 to 255 (unsigned).
- Saves memory and speeds up calculations, but less detail.



int16

- Uses 16 bits, storing 65,536 possible values.
- Example: Range can be -32,768 to 32,767 (signed).
- Uses more memory, but offers higher accuracy and detail.



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With or Without Clamping

What is Clamping?

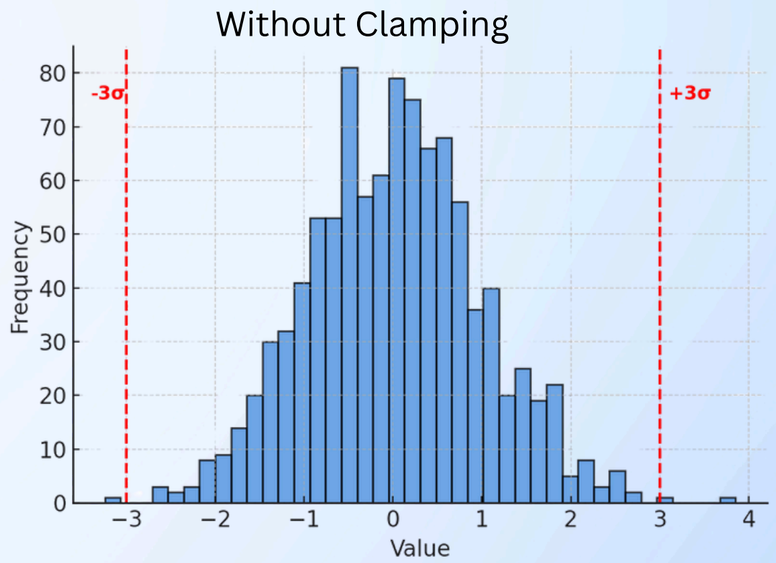
Restricts values to stay within a fixed range (min–max), improving quantization accuracy.

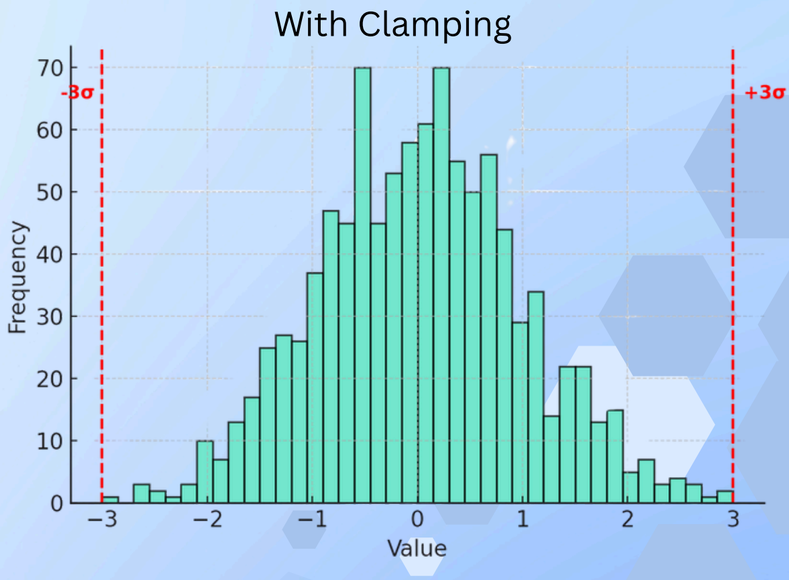
Methods We Tested:

Standard Deviation: 3σ / 4σ

Percentages: 0.1% / 1%

Example for clamping 3σ





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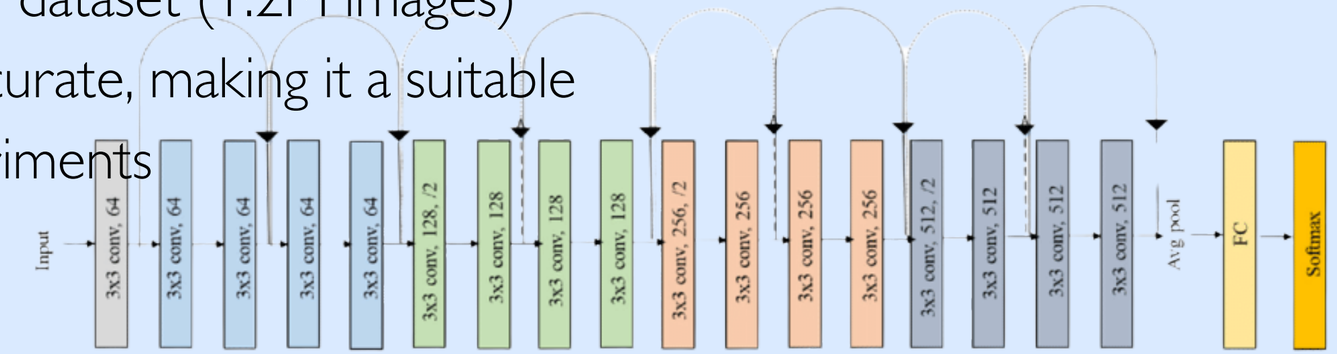
Dataset: (from "Animals-10 dataset")

- Contains 10 animal categories.
- Includes ~100 sampled images.



Model: ResNet18

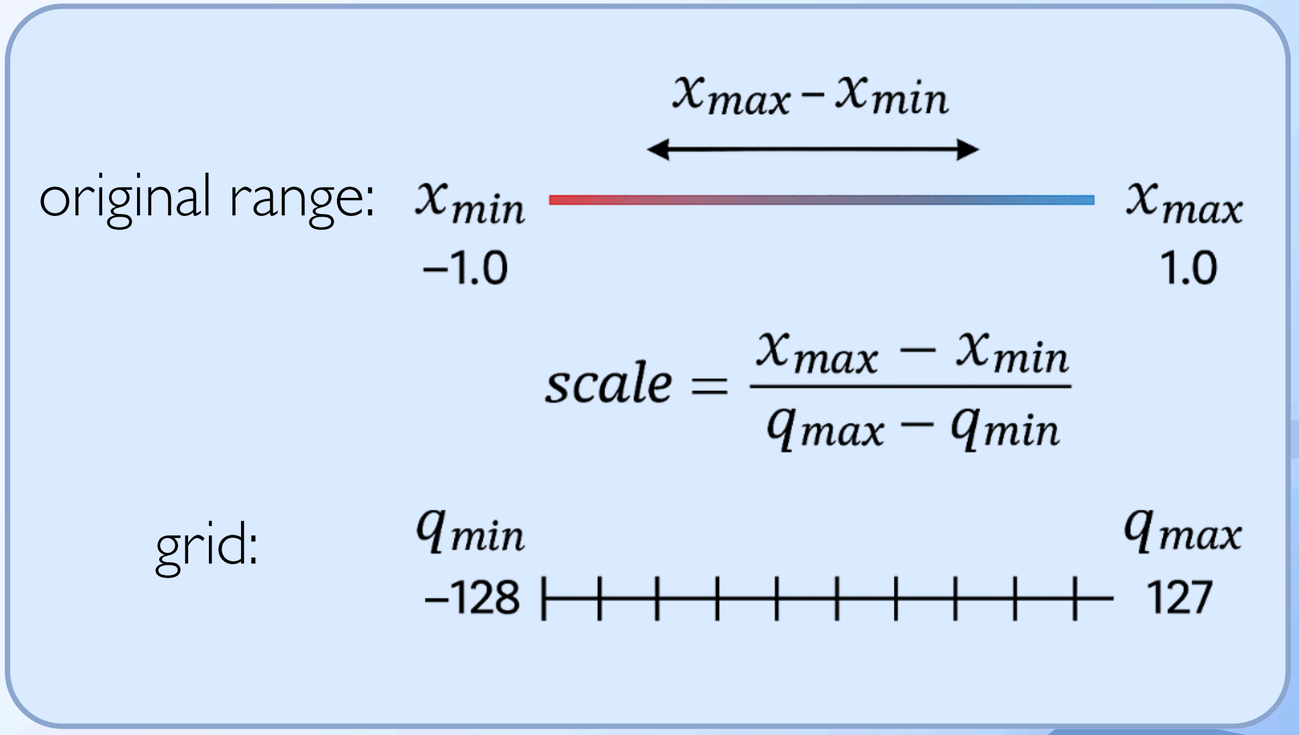
- ResNet18 is a convolutional neural network with 18 layers
- Pre-trained on the "ImageNet" dataset (1.2M images)
- ResNet18 is lightweight yet accurate, making it a suitable baseline for quantization experiments



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The quantization process

In the quantization process, the **scale** determines how large each quantization step is.
Each integer value represents a range of original float values.
It affects how real-valued numbers are mapped into discrete integer levels.



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Quantization Mapping

Asymmetric Quantization

- Values are not centered around 0.
- Zero point is added to shift the range

$$\text{round}\left(\frac{x}{\text{scale}} + \text{zero_point}\right) = q$$

Symmetric Quantization

- Values are centered around 0.
- No zero point is needed.

$$\text{round}\left(\frac{x}{\text{scale}}\right) = q$$

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Quantized Equation

$$\hat{\mathbf{X}} = s_{\mathbf{X}} \cdot \mathbf{X}_{\text{int}} \approx \mathbf{X}$$

X - Original real value (before quantization)

x_{int}- Quantized integer value (discrete representation)

s_x- Scale factor, the constant used to map between integer and real domains

$\hat{x}$- Reconstructed (dequantized) value, an approximation of the original real number

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Quantized Neural Network Equation

- $\hat{A}_n$ - Quantized accumulator output
- $\hat{b}_n$ - Bias
- $\hat{W}_{n,m}$ - Quantized weight
- $\hat{x}_m$ - Quantized activation (input)
- S_x , S_w -Scale factors for weights & activations
- W^{int}, x^{int} - Integer representations

$$\hat{A}_n = \hat{\mathbf{b}}_n + \sum_m \widehat{\mathbf{W}}_{n,m}\,\hat{\mathbf{x}}_m$$

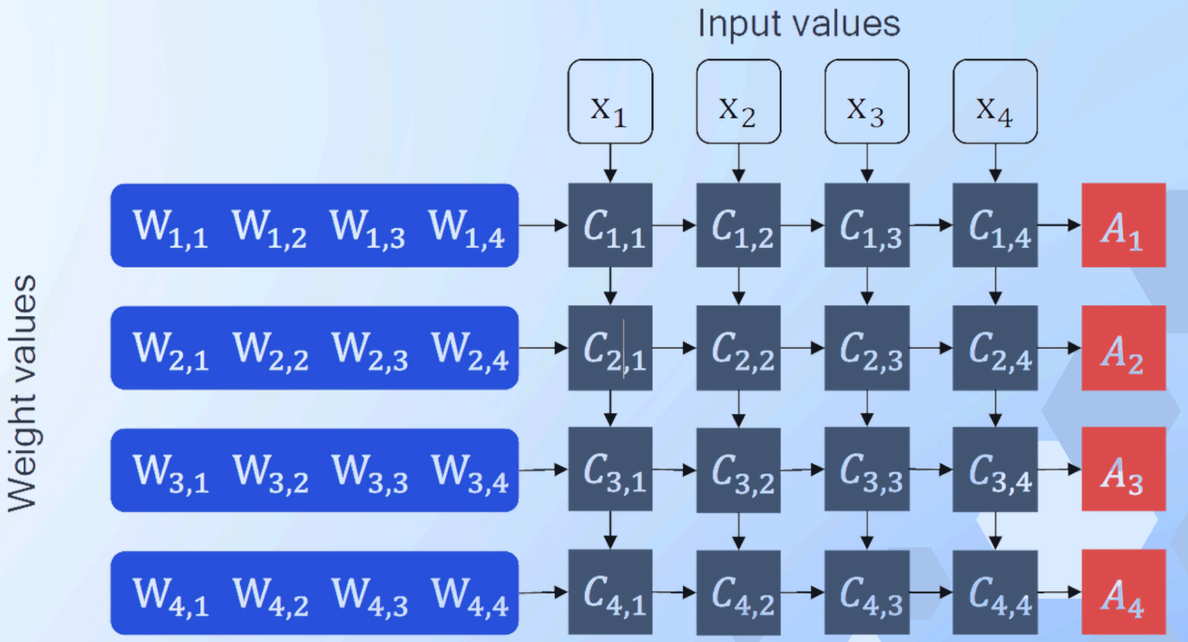
$$= \hat{\mathbf{b}}_n + \sum_m \left(s_{\mathbf{w}}\mathbf{W}^{\text{int}}_{n,m}\right)\left(s_{\mathbf{x}}\mathbf{x}^{\text{int}}_m\right)$$

$$= \hat{\mathbf{b}}_n + s_{\mathbf{w}}s_{\mathbf{x}}\sum \mathbf{W}^{\text{int}}_{n,m}\,\mathbf{x}^{\text{int}}_m$$

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Matrix Multiplication in Neural Networks

- $W_{i,j}$ – Weight values (matrix entries)
- X_j – Input activations (vector)
- $C_{i,j}$ – Multiply results
- A_i – Accumulators (sum of products for each row)
- Process: Multiply inputs × weights → Accumulate → Output



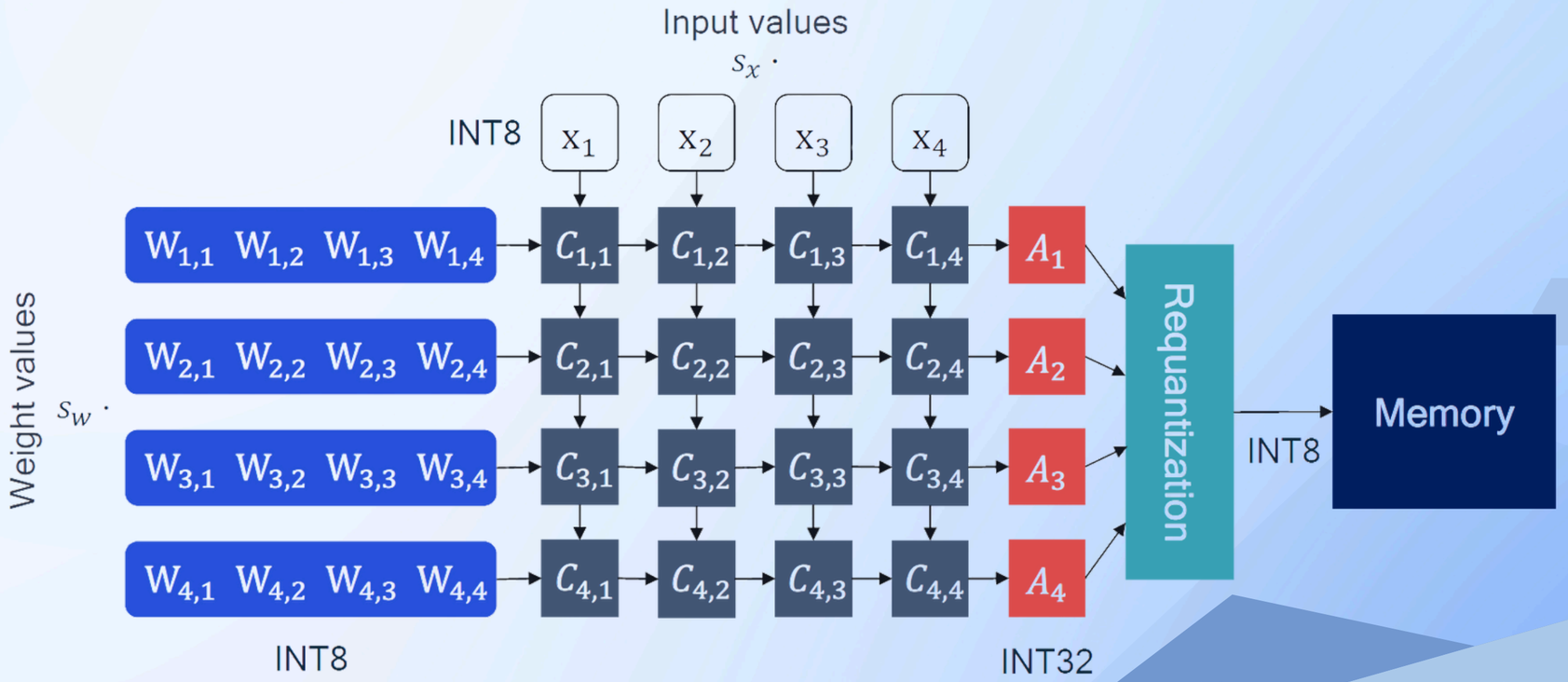
Multiply → Accumulate → Output

*A White Paper on Neural Network Quantization

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Quantized Matrix Multiplication

- Weights and inputs stored as INT8
- Multiplications and additions performed as MAC operations
- Results collected in 32-bit Accumulators
- Requantization step → back to INT8
- Reduced memory transfer and energy cost



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Quantized Matrix Multiplication

Accuarcy

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Measures classification correctness on the test set (percentage of correct predictions).

MSE (Mean Squared Error)

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Measures reconstruction error, penalizing larger errors more strongly.

MAE (Mean Absolute Error)

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Measures the average absolute difference between original and quantized values.

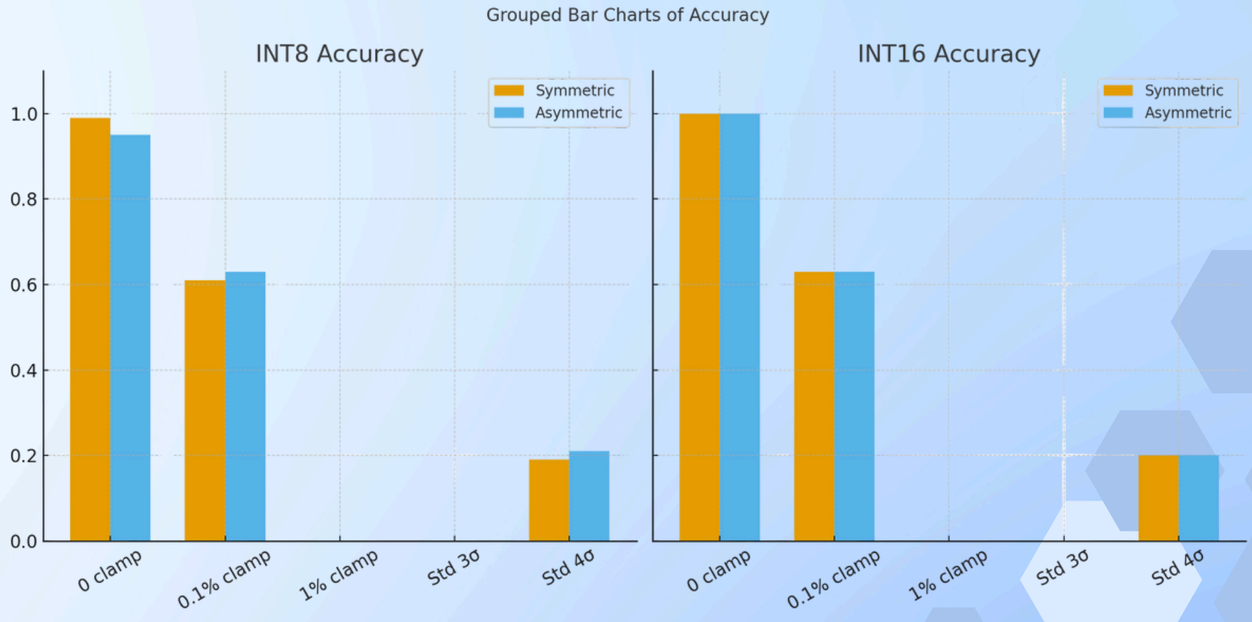
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Results -Accuarcy

INT8: Achieved the highest accuracy without clamping, with symmetric quantization performing slightly better.
INT16: Showed high accuracy and minimal difference between symmetric and asymmetric quantization.

Conclusion:

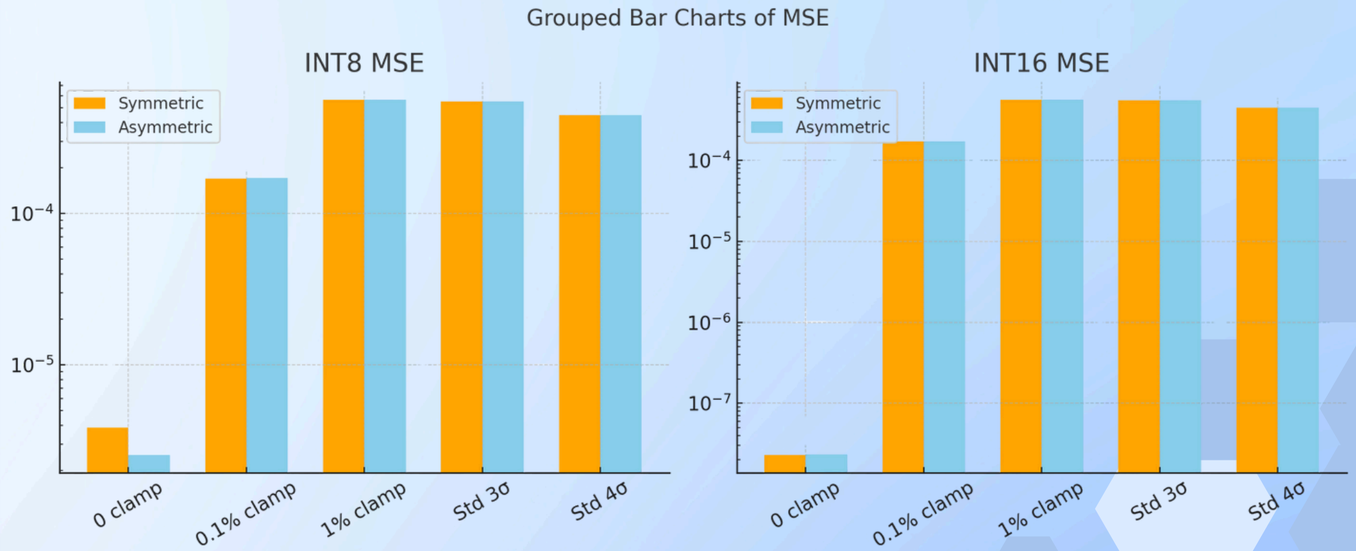
- Clamping generally decreases accuracy.
- INT16 provides higher and more stable overall accuracy



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Results- MSE

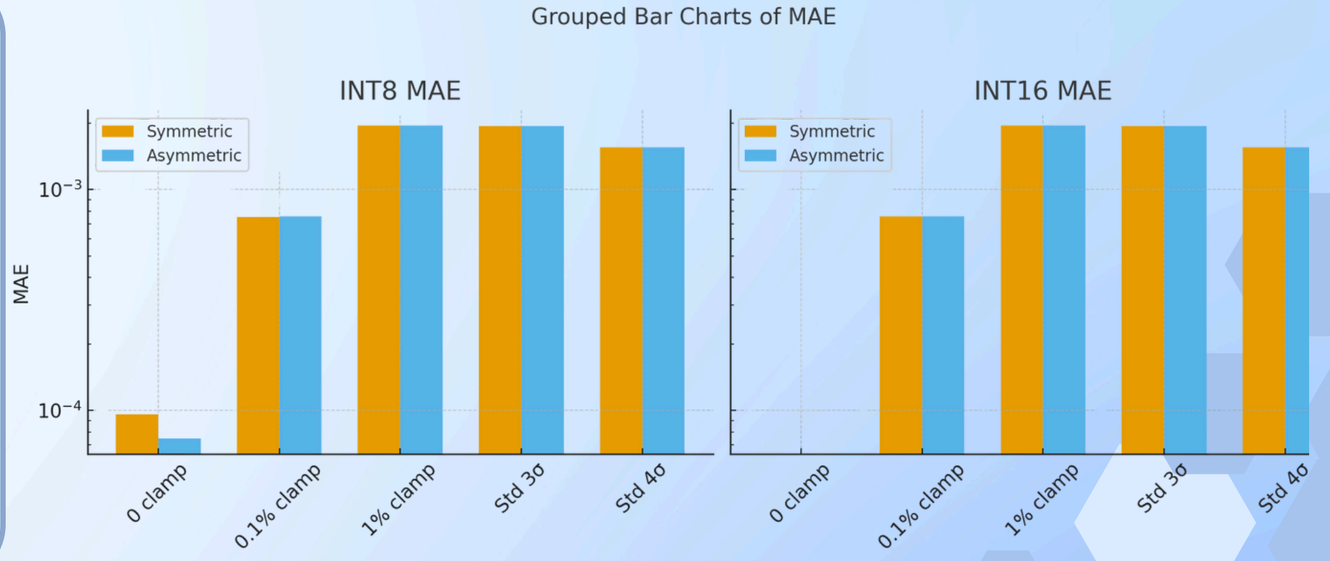
- Lowest error at 0 clamp for both INT8 and INT16
- Clamping increases MSE across all strategies
- INT16 MSE < INT8 MSE (more precise)
- Symmetric ≈ Asymmetric (almost identical error)



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Results- MAE

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Conclusions

1. **INT16** achieves the highest accuracy (100% without clamping).
2. **INT8** offers a strong trade-off: ~99% accuracy with lower memory/compute.
3. **Clamping** generally reduces accuracy and should be avoided.
4. **Symmetric** ≈ **Asymmetric**, with only minor differences in special cases
5. **Best trade-off: INT16 → accuracy🎯 | INT8 → efficiency⚡**

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Future Work

Our configuration works well but may not be optimal for all models or edge cases

Future direction:

- Develop an adaptive quantization framework that:
- Learns from diverse data and model types.
- Dynamically selects the best quantization strategy per model/dataset.
- Maximizes both accuracy and efficiency
- Ensures robust performance in real-world scenarios.

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